



AHMED H. HANFY

Mechanical Engineer ||
Aerodynamic specialist

PROFILE

Doctoral researcher in the Marie Curie fellowship program with expertise in experimental aerodynamics. Proficient in Python, MATLAB, and C++; skilled in data/image analysis. Proficient in CAD modeling using Autodesk Inventor and adept with Siemens NX. Seeking to contribute expertise, pursue learning, and advance career.

CONTACT ME

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LANGUAGES

- English
- Italian
- Polish
- Arabic

SKILLS

- Ansys (Fluent)
- Inventor
- Siemens NX
- Python
- MATLAB
- LabVIEW

OTHER SKILLS

Wind-tunnel operation | Test planning | Research | Manufacturing | 3D printing | Technical drawings | C++ | Fortran | OpenMPI | MATLAB GUI | PCA | Analysis | Machine learning | Image analysis | Signal processing | OOP | MS Office.

ACTIVITIES

Summer trainings [Hydro-electric stations, Diesel engines & Hydraulic maintains (MANTRAC), ASME CFD workshop].
Volunteering [researcher's night, Science Club Chairman, Torpedo robotics team: Participate in the MATE ROV competitions to perform underwater tasks].

EXPERIENCE

- Doctoral Researcher (Aerodynamic specialist) [11/2020 - Present]
Institute of fluid-flow machinery polish academy of sciences, Gdansk (Poland)
Full-time employment at IMP PAN transonic wind tunnel as an experimental aerodynamics specialist, played a key role in various projects:
- **SMS Project (Vibrating trailing edge of a morphing supercritical airfoil)**
 - Conducted a POD analysis on PIV data for the Airbus A320 aerofoil in the transonic regime, effectively eliminating noise from the velocity dataset.
 - Executed precise analysis of velocity fluctuations in the wake.
 - **TEAMAero Project (SBLI on compressor rotor profile with flow control)**
 - Led test planning for experimental investigations into the effects of manufacturing and roughness on the transonic compressor rotor/fan.
 - Coordinated collaboration with project partner Rolls Royce Deutschland.
 - Modified the design and provided detailed drawings for an existing single-passage fan profile.
 - Operated a blowdown wind tunnel.
 - Implemented various measurement techniques to capture aerodynamic performance accurately.
 - Engineered a 30% increase in suction efficiency by modeling the piping system to control corner flows.
 - Enhanced the wind tunnel feedback system and fast camera triggering using LabVIEW and DAQ devices, achieving a remarkable accuracy of 0.6ms.
 - Developed Python tools and packages for image/signal processing, test data, and analysis, significantly enhancing understanding of flow physics.

SCIENTIFIC VISITS AND INTERNSHIPS

- Visiting Researcher [08/2022 - 09/2022]
German Aerospace Centre (DLR), Cologne (Germany)
Within Marie Skłodowska-Curie Actions, a scientific secondment was made to DLR Transonic Cascade Wind Tunnel, successfully executed unsteady measurements, including:
- Calibration of Kulite probe and vibration sensors.
 - Application analysis on high Data Acquisition rates.
 - Synchronization of the speed Schlieren.

- Research Internship [02/2020 - 05/2020]
Institute of fluid-flow machinery polish academy of sciences, Gdansk (Poland)
As a complementary course during master's, accomplished the following objectives:
- Proficiently extracted quantitative information from interferograms for experimental study relevant to SWBLI.
 - Developed a MATLAB GUI involving FFT and phase shifting.
 - Improved phase detection accuracy through the application of machine learning techniques, including DBSCAN, Linear regression, etc.

EDUCATION

- Ph.D. Mechanical engineering [11/2020 - Present]
Marie Skłodowska-Curie Actions, Innovative Training Networks HORIZON 2020
Institute of fluid-flow machinery polish academy of sciences, Poland
(Est. Graduation, late 2024)
- M.Sc. Applied Mathematics and Mathematical Engineering
InterMaths Joint MSc Programme [09/2018 - 05/2020]
 - Ivan Franko National University of Lviv (IFNUL), Ukraine [2019 - 2020]
 - University of L'Aquila (UAQ), Italy [2018 - 2019]
- B.Sc. Mechanical Engineering [09/2011 - 07/2016]
Alexandria University (AU), Egypt

RECENT PUBLICATIONS

- Hanfy, A. H., et al., *Advancements in Shock-Wave Analysis and Tracking from Schlieren Imaging*. Measurements Journal. (under review).
- Nel, P. L., et al (2024). *Effect Of Transition On Self-sustained Shock Oscillations In Highly Loaded Transonic Rotors*. AIAA Journal, 0(0):1–13
- Nel, P. L., et al (2023). *Towards Resolving Natural Shock Oscillation and Mitigation of Altitude Excitation in a Transonic Fan*. In AIAA AVIATION 2023 Forum (p. 4001).

SELECTED PROJECTS

- Shock Tracking Library (OpenCV, Python)
- Finite Fringe Analysis for Optical Measurement of Compressible Fluid Flow Parameters (MATLAB GUI application - MSc. Thesis)
- Parallel implementation of Poisson's equation (OpenMPI, Fortran and C++)



Full CV